

Yizhe Zhu

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🐙 Github 🌐 Personal Website 🎓 Scholar

EDUCATION

Northeastern University
M.S. in Robotics, GPA: 3.8/4.0

Boston, MA, USA
Sep. 2024 – Present

Xi'an Jiaotong Liverpool University
B.Eng. in Mechatronic and Robotic Systems, GPA: 3.6/4.0 (Rank: 4/38)

Suzhou, China
Sep. 2020 – Jul. 2024

PUBLICATIONS

- **Yizhe Zhu**, Zhang Ye, Boce Hu, Haibo Zhao, Yu Qi, Dian Wang, Robert Platt. “Residual Rotation Correction using Tactile Equivariance.” *Under review at IEEE Robotics and Automation Letters (RA-L)*, 2025. [\[Link\]](#)
- Xupeng Zhu, **Yizhe Zhu***, Yu Qi*, Robin Walters, Robert Platt. “EquAct: An SE(3)-Equivariant Multi-Task Transformer for Open-Loop Robotic Manipulation.” *Under review at International Conference on Learning Representations (ICLR)*, 2026. [\[Link\]](#)
- Haibo Zhao*, Dian Wang*, **Yizhe Zhu**, Xupeng Zhu, et al. “Hierarchical Equivariant Policy via Frame Transfer.” *International Conference on Machine Learning (ICML)*, 2025. [\[Link\]](#)
- Haibo Zhao, Yu Qi, Boce Hu, **Yizhe Zhu**, Ziyang Chen, et al. “Generalizable Hierarchical Skill Learning via Object-Centric Representation.” *Under review at IEEE Robotics and Automation Letters (RA-L)*, 2025. [\[Link\]](#)
- Yu Qi, Haibo Zhao, Ziyu Guo, et al., **Yizhe Zhu**, et al. “BEAR: Benchmarking and Enhancing Multimodal Language Models for Atomic Embodied Capabilities.” *Under review at International Conference on Learning Representations (ICLR)*, 2026. [\[Link\]](#)

RESEARCH EXPERIENCE

Contact-Rich Robust Manipulation *Boston, MA, USA*
Helping Hands Lab, Northeastern University *Aug. 2025 – Present*

- Developing robust manipulation policies through integration of Diffusion Policy Optimization (DPPO) and residual correction networks to improve success rates in fine manipulation tasks.
- Developing hybrid frameworks combining diffusion-based policy learning with closed-loop tactile feedback for high-precision manipulation.

Tactile-Based Rotation Residual Estimation *Boston, MA, USA*
Helping Hands Lab, Northeastern University *Mar. 2025 – Aug. 2025*

- Developed EquiTac, an SO(2)-equivariant framework for real-time rotation residual estimation using GelSight sensors, leveraging geometric symmetries to enhance sample efficiency.
- Developed an integrated training pipeline that combines tactile tracking with closed-loop control, achieving improved generalization across diverse object geometries.

Teleoperation Data Collection System *Boston, MA, USA*
Helping Hands Lab, Northeastern University *Oct. 2024 – Mar. 2025*

- Designed a teleoperation system using 3D SpaceMouse for synchronized collection of closed-loop and open-loop demonstration data.
- Developed a multi-modal data acquisition system for synchronous recording of action trajectories, visual observations, and robot states to facilitate imitation learning.

YOLO-Based Visual Tracking Car *Suzhou, China*
Xi'an Jiaotong Liverpool University *Dec. 2023 – Apr. 2024*

- Trained YOLO model for real-time QR code detection with binocular camera for depth estimation and autonomous vehicle tracking control.
- Developed Python-Arduino serial communication protocols ensuring low-latency real-time control for mobile robot navigation.

HONORS AND AWARDS

- XJTLU Academic Scholarship, University Academic Great Award
- ABB Cup Innovation Contest, First Prize of Motion Control Competition

TECHNICAL SKILLS

Programming Languages: Python, C/C++, MATLAB
Robotics & ML Frameworks: PyTorch, OpenCV, ROS, MuJoCo, Robomimic, RLbench
Hardware Experience: GelSight Tactile Sensors, UR5 Robot Arms, Realsense Cameras, 3D Mouse, Arduino
Design & Modeling: Blender, 3D Modeling, 3D Printing, Photoshop

RESEARCH INTERESTS

Robot manipulation combining imitation learning and reinforcement learning; Human-in-the-loop continual learning for robotics; Tactile sensing and closed-loop control for contact-rich tasks;